

Kunal Jangid

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🔗 kunaljangid01 📄 Google Scholar

About Me!

I am a PhD scholar at IISER Bhopal, working on reliable tumor and OAR segmentation for radiotherapy planning using multi-modal medical imaging (CT/MRI/PET). My research focuses on improving boundary precision and clinical reliability under real deployment constraints such as limited labeled data, acquisition anisotropy, inter-observer variability, and cross-site domain shift. I build methods that combine foundation-model priors with efficient 3D segmentation backbones and uncertainty/calibration to support clinician-in-the-loop review. I also have ongoing **collaborations** with **AIIMS Bhopal** and **JNCH Bhopal** to validate models in resource-constrained hospital environments. My goal is to develop segmentation systems that are both robust in practice and **deployable on resource-constrained clinical workstations**.

Education

Ph.D Scholar

Supervisor - Dr. Vinod K. Kurmi

Dept. of Data Science and Engineering, Indian Institute of Science Education and Research, Bhopal

Topic: **Reliable Tumor Delineation from Multi-Modal Medical Imaging for Precise Radiotherapy Planning.**

Coursework: Deep Learning, Biometrics: An Introduction to Research, Advanced Deep Learning, 3D Deep Learning and Applications, Data Science in Practice, Natural Language Processing

July 2024 – Present

CGPA: 9/10

Integrated M.Sc. in Computer Science

Dept. of Computer Science, Central University of Rajasthan

BSc: **75.8%** and MSc: **85.6%**

July 2018 – June 2023

CGPA: 8.04/10

Intermediate

Demonstration Multipurpose School (RIE), Ajmer

2018

79% (CBSE)

Matriculation

Emmanuel Public School, Kishangarh

2016

CGPA: 8.8/10 (CBSE)

Experience

Project JRF

PI - Dr. Vinod K. Kurmi

Dept. of Data Science and Engineering, Indian Institute of Science Education and Research, Bhopal

Sept 2023 – July 2024

- Worked on a collaborative project between IISER Bhopal and AIIMS Bhopal, handling complex 3D DICOM datasets (CT/MRI).
- Developed "SpectSeg," a spectral-domain segmentation model to handle high-frequency noise in low-dose CT scans.
- Collaborated with clinical partners to curate datasets, focusing on the standardization of heterogeneous oncology data.

Website Developer

Unvoiced Media & Entertainment Pvt. Ltd.

Develop the website of the company and maintain it. Video editing of the open mic events.

Dec 2021 – Dec 2022

Patent

Tanmay Basu, Vinod K Kurmi, Prateek Sarangi, **Kunal Jangid**, Rajesh Pasricha, Riya Agarwal, Avaneesh Mishra **"IMPACT: Automatic Delineation of Tumor and Organs for Precise Radiotherapy Planning in Head and Neck Cancer"**, India (In Progress)

2025

Publications

Kaushal, A., **Jangid, K.**, Kurmi, V. K. (2025). **CLFseg: A fuzzy-logic based solution for boundary clarity and uncertainty reduction in medical image segmentation**. In Proceedings of the 36th British Machine Vision Conference (BMVC 2025). British Machine Vision Association.

BMVC 2025 Conference
(Core A)

[Paper Link](#)

Rai, S., Thakur, R. S., **Jangid, K.**, & Kurmi, V. K. (2025, February). **Label Calibration in Source Free Domain Adaptation**. In 2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) (pp. 6446-6455). IEEE.

WACV 2025 Conference
(Core A)

[Paper Link](#)

Meena, G., Indian, A., Mohbey, K. K., & **Jangid, K.** (2024). **Point of interest recommendation system using sentiment analysis**. Journal of Information Science Theory and Practice, 12(2), 64-78.

2024 Journal

[Paper Link](#)

Paper Under Submission

ReLiF-3D: Robust Lesion-Aware and Foundation-Guided Semi-Supervised 3D Medical Image Segmentation

CVPR 2026

Kunal Jangid, Tanmay Basu, Vinod K. Kurmi

SpectSeg: Frequency-Aware Transformer for Precise Boundary Delineation in Medical Image Segmentation

Journal

Kunal Jangid, Tanmay Basu, Dr. Rajesh Pasricha, Vinod K. Kurmi

Mitigating Domain Shift in 3D Black-box Domain Adaptation through Prototypical Alignment

Journal

Kunal Jangid, Shivangi Rai, Vaibhav Kumar, Vinod K. Kurmi

Projects

Medical Image Segmentation of Tumor and Organs

IISER Bhopal

Developed an automated segmentation model to segment the anatomical structures of the tumors from the head & neck CT scan (in-house) dataset and multiple organs from the public dataset.

Sep 2023 - Jul 2024

POI Recommendation System using Sentiment Analysis

CURAJ, Jaipur

Develop a recommendation system that incorporates the sentiments of user reviews, user location, and user interests to provide accurate and relevant recommendations. The recommendation system will consider the user's location, interests, and recommend POIs that match their interests and have positive or neutral sentiment based on user reviews.

Jan 2023 - Jul 2024

POI recommendation based on Adversarial Learning in LBSN

CURAJ, Jaipur

Goal is to recommend the Points of Interest (POI) to users that the user is likely to visit based on their location, check-in history and other contextual information in Location-Based Social Networks (LBSN). The project aimed to improve the accuracy and robustness of POI recommendations by using adversarial learning techniques.

Jul 2022 - Dec 2022

Neural Style Transfer with VGG

CURAJ, Jaipur

The goal of the project was to transfer the style of a particular artwork onto another image using the VGG convolutional neural network architecture.

Jan 2022 - May 2022

Twitter Sentiment Analysis using Machine Learning

The aim is to develop a model that trains on the sentiment of tweets from Twitter using machine learning algorithms and classifies the sentiment of tweets collected from Twitter in real-time.

CURAJ, Jaipur
Mar 2021 - Aug 2021

Technologies

- **Core Areas:** Medical Image Analysis, 3D Computer Vision, Segmentation, Semi/Unsupervised Learning.
- **Deep Learning:** PyTorch, TensorFlow, Transformers (ViT, UNETR), Foundation Models (SAM-Med3D, MedSAM).
- **Medical Tools:** Monai, Nibabel, SimpleITK, 3D Slicer, DICOM standards.
- **Programming:** Python, C, C++, Java, C#, Linux/Bash Scripting, JavaScript, ReactJS, $\LaTeX$, MySQL.

Achievements

- GATE qualified in CS (2023)
- UGC NET-LS qualified in CS (2023)
- Received Grant from TiiC (TDK India Incubation Center), GDC, IIT Madras to attend the TiiC Accelerator Workshop 2024 at IIT Madras
- Received Grant from IndoML 2024 Travel Grant to attend the IndoML 2024 at BITS Pilani Goa.

Workshops Attended

- "Brain & Computing: Current Trends and Future Prospect" - One day (14th February 2020) workshop organized by the Department of Computer Science at Central University of Rajasthan
- "Securing the Smart Cities with Internet of Things (IoT) and Blockchain" - 3 days (5-7 March, 2020) workshop organized by the Department of Computer Science at Central University of Rajasthan
- "Deep Learning in Image Processing and Pattern Recognition" - Two-Week (March 13-22, 2023) Faculty Development Program organized by the Department of Computer Science at Central University of Rajasthan
- Symposium on AI in Healthcare on 22th March 2025 organized by Dept. of Data Science and Engineering, IISER Bhopal
- "Microsoft Powered One Day Power Workshop on AI Innovation" - on 21st June 2025, hosted by FITT, IIT Delhi, and Dept. of Data Science and Engineering, IISER Bhopal

Teaching Assistant

Data Structures and Algorithms

*Jan 2026- April 2026
(Present)*

Fundamentals of Database Systems

*Aug 2024 - Nov 2024
& Aug 2025 - Nov 2025*

Introduction to Programming

Jan 2025 - April 2025